

MIAE 221 · Rapid Review Sheet · Part 6

Mechanical Properties of Metals

1. Stress, Strain & Elasticity (Lecture 10)

- **Loading modes:** tension, compression (σ and ϵ negative by convention), shear, torsion (shafts, axles, drills).
- **Engineering Stress:** $\sigma = \frac{F}{A_0}$, shear $\tau = \frac{F_s}{A_0}$ (MPa = N/mm²). Original area A_0 ; actual area gives true stress.
- **Engineering Strain:** $\epsilon = \frac{\Delta l}{l_0}$, lateral $\epsilon_L = \frac{-\delta_L}{w_0}$, shear $\gamma = \tan \theta$. Always dimensionless.
- **Hooke's Law:** $\sigma = E\epsilon \iff \frac{F}{A_0} = E \frac{\Delta l}{l_0}$. E = stiffness = elastic slope: W, Ta, Mo **steep**; Al, Cu, Ag **shallow**. Ceramics 300, steel 207, Cu 110, plastics 3 GPa. Slide 11 wire: $E = \frac{17.24/0.55 \text{ mm}^2}{1.68/10\,000} \approx 187 \text{ GPa}$.
- **Atomic Origin of E :** $E \propto \left(\frac{dF}{dr}\right)_{r_0}$. Strong bonds $\implies$ high E , high T_m , low α . E falls as T rises: thermal expansion moves atoms to a less steep part of the force curve.
- **Non-linear elastic** (cast iron, concrete, some polymers): **tangent** modulus = slope of tangent at σ ; **secant** modulus = slope of line from origin to σ .
- **Poisson's Ratio:** $\nu = -\frac{\epsilon_x}{\epsilon_z} = -\frac{\Delta d/d_0}{\Delta l/l_0}$ (0.2–0.5; metals $\approx$ 0.3). **Isotropic:** $E = 2G(1 + \nu)$, $\tau = G\gamma$ (not for composites or single crystals).
- **Anelasticity:** time-dependent but fully recoverable elastic strain; small in metals, significant in polymers (viscoelastic).

2. Plastic Deformation, Yielding & Tensile Strength

- **Yield Strength (σ_y):** Standardized by the 0.002 (0.2%) **strain offset** line parallel to E : $\sigma = E(\epsilon - 0.002)$.
- **Upper & Lower Yield Point:** In annealed low-carbon steel due to interstitial carbon/nitrogen pinning (Cottrell atmospheres $\rightarrow$ Lüders bands). Lower yield is used for design.
- **Ultimate Tensile Strength (σ_{UTS}):** Maximum engineering stress on curve: $\sigma_{UTS} = \frac{F_{\max}}{A_0}$. Marks the onset of **necking**.

3. Ductility, Resilience & Toughness

- **Percent Elongation:** $\%EL = \left(\frac{l_f - l_0}{l_0}\right) \times 100\%$
- **Percent Reduction in Area:** $\%RA = \left(\frac{A_0 - A_f}{A_0}\right) \times 100\% = \left(1 - \frac{d_f^2}{d_0^2}\right) \times 100\%$ (square diameters!).
- **Modulus of Resilience (U_r):** Elastic energy absorbed up to yielding:

$$U_r = \frac{1}{2}\sigma_y\epsilon_y = \frac{\sigma_y^2}{2E} \quad (\text{J/m}^3 \text{ or kJ/m}^3)$$

- **Toughness:** Total area under the complete $\sigma - \epsilon$ curve up to fracture. Requires combination of strength AND ductility.

4. True Stress-Strain & Strain Hardening

- **True Stress:** $\sigma_T = \frac{F}{A_i} = \sigma(1 + \epsilon)$ (valid prior to necking).
- **True Strain:** $\epsilon_T = \ln\left(\frac{l_i}{l_0}\right) = \ln(1 + \epsilon)$ (valid prior to necking).
- **Hollomon Power Law:** $\sigma_T = K \cdot \epsilon_T^n$ (n = strain hardening exponent, 0.10 – 0.50).
- **Necking Criterion:** At UTS, $\frac{d\sigma_T}{d\epsilon_T} = \sigma_T \implies \epsilon_T = n$.

5. Hardness Testing Reference

- **Brinell (HB):** Steel/tungsten ball ($D = 10 \text{ mm}$):

$$HB = \frac{2P}{\pi D (D - \sqrt{D^2 - d^2})}$$

- **Rockwell:** Diamond cone (120°) or steel ball (1/16"). Read depth directly: **HRA** (cemented carbides), **HRB** (brass/Al), **HRC** (hardened steels).
- **Vickers (HV):** Diamond pyramid (136°). Micro/macro hardness on a single scale.
- **Knoop (HK):** Microhardness for thin coatings and brittle ceramics.
- **Empirical Correlation:** $TS \text{ (MPa)} \approx 3.45 \times HB$.